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Base Models Can Reason By Taking a Cue From Training Data

A Single Starting Token Can Raise Math Accuracy by 36 pp in a Base Model—and the Effect Is Causally Traceable to Training Data, Not Model Architecture

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Base language models harbor latent reasoning competence that is largely dormant until activated by a specific starting token—a *reasoning cue* whose effect is not semantic but distributional, mechanically traceable to the cluster of step-by-step reasoning documents in the training corpus that bear the same token.

AT A GLANCE

Problem: Reinforcement learning post-training is costly; the mechanism by which it elicits reasoning chains is poorly understood.

Why it matters: If RL merely selects regions of a base model’s training distribution, cheaper interventions should reproduce the gains.

Method: Systematic sweep over starting-token cues; causal data interventions to add or remove reasoning associations.

Result: The cue “.\n\nOkay” raises Olmo-3-7B MATH-500 from 42% to 78%; causal interventions confirm the training-data mechanism.

Evidence: Evaluated on MATH-500, coding benchmarks, and multiple frontier base models; causal interventions with data ablations.

The Big Picture

Large language models trained with reinforcement learning from verifiable rewards (RLVR) achieve dramatic performance gains on mathematical reasoning and code generation. Yet the precise mechanism has remained unclear: does RL teach the model new reasoning skills, or does it teach the model *when* to deploy skills that the base model already possesses?

This paper proposes a third, more concrete answer: RL teaches the model to begin its response with particular starting tokens—reasoning cues—that route generation

toward the region of the training distribution in which careful step-by-step reasoning documents reside. The paper provides both correlational and causal evidence for this mechanism.

Why Existing Approaches Fall Short

RL post-training is expensive. RLVR requires generating many rollouts and computing rewards, multiplying training compute relative to supervised fine-tuning. If the mechanism is a distributional routing effect, a cheaper intervention should suffice.

Prompt engineering is empirical. Prior chain-of-thought work shows that phrases like “Let’s think step by step” improve performance, but attributes the effect to semantic instruction following—a hypothesis that predicts the effect should generalize to semantically equivalent phrases, which the paper tests and finds false.

Base model latent capacity is underutilized. The default starting token chosen by a practitioner is arbitrary; this work shows it has outsized influence on the quality of the continuation.

Core Mechanism

A **reasoning cue** is a short prefix c placed at the start of a base model’s response. The central finding is that effective cues are not semantically special: their power comes from co-occurring in the training corpus with step-by-step reasoning chains, not from the meaning of the words themselves.

When a cue c appears at the start of the model’s response, next-token prediction conditions on c and selects continuations from the region of the training data distribution where c was associated with careful, multi-step reasoning. This makes the cue a *distributional selector* rather than a semantic instruction.

Technical Formulation

Let $\mathcal{M}$ be a base LLM with vocabulary $\mathcal{V}$. For a problem q and cue string $c \in \mathcal{V}^*$:

$$\text{Acc}(c) = \mathbb{E}_q[\text{Correct}(\mathcal{M}(q, c))]$$

The sweep identifies the optimal cue $c^* = \arg \max_c \text{Acc}(c)$.

For the causal intervention, let $D_c \subseteq \mathcal{D}_{\text{train}}$ be training documents whose response begins with c . The paper shows:

$$\text{Acc}_{D \setminus D_c}(c) \ll \text{Acc}_D(c)$$

Conversely, augmenting the training data with arbitrary-token-prefixed reasoning traces $\tilde{D}_c$ creates a new effective cue:

$$\text{Acc}_{D \cup \tilde{D}_c}(c') \approx \text{Acc}_D(c^*)$$

Learning or Inference Procedure

Finding effective cues:

1. Enumerate a large set of candidate starting tokens and phrases.
2. For each cue, generate responses on a held-out problem set.
3. Score responses and rank cues by mean accuracy.

Causal confirmation:

1. Identify training documents co-occurring with the effective cue.
2. Remove those documents and retrain (or fine-tune) a model.
3. Verify that the cue’s effect disappears with the associated data.

Creating custom cues:

1. Collect reasoning traces; prepend an arbitrary token to each.
2. Fine-tune on these modified traces.
3. The arbitrary token becomes an effective cue.

What the Guarantee Says

The guarantee is empirical: across all tested base models and cue candidates, the reasoning benefit of any cue is causally traceable to its associated training documents. Removing those documents degrades the cue; adding documents that associate an arbitrary token with reasoning creates a new effective cue. The mechanism is not model-specific—it holds for Olmo-3-7B, Qwen3-14B, and several other frontier base models.

Experimental Findings

- **Olmo-3-7B:** the cue “.\n\nOkay” raises MATH-500 pass@1 from **42% to 78%**, matching the RL-trained counterpart.
- **Qwen3-14B:** “Alright,” raises MATH-500 from **72% to 87%**, also competitive with RL.
- **Code generation:** analogous cues improve pass@*k* metrics on coding benchmarks.
- **Causal intervention:** removing documents associated with the effective cue eliminates its accuracy benefit.
- **Custom cue creation:** the word “chicken,” after fine-tuning on reasoning traces prefixed with it, becomes an effective cue.

Ablations and Interpretation

Semantic vs. distributional hypothesis: Replacing the effective cue with semantically similar phrases that

do not appear in the relevant training data produces no reasoning gain, confirming the distributional mechanism.

Context length stability: The cue effect holds even with few-shot examples in context, ruling out attention-dilution as an alternative explanation.

Multiple cues: Combining multiple effective cues does not additively compound gains, indicating they tap overlapping training data regions.

Implications for RL: Since base-model cue performance matches RLVR, the paper argues that RL primarily teaches models to start responses in the step-by-step reasoning region of their training distribution—not to acquire fundamentally new reasoning capabilities.

Reference

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